

Phospholemman, also known as FXYD1, is a single-pass transmembrane protein that regulates the Na^+, K^+ ATPase pump. Phospholemman consists of four alpha helices: H1 through H4. H1 is located on the extracellular side of the cell. H2 forms the transmembrane region of the protein and interacts intimately with helix TM9 of the Na^+, K^+ ATPase pump. H3 and H4 face the intracellular side of the membrane, where H4 interacts with the cytosolic domain of the Na^+, K^+ ATPase pump through electrostatic interactions. Within H4 exist multiple phosphorylation sites, including S68 and S63, which are phosphorylated by protein kinase A (PKA) and protein kinase C (PKC), respectively. Phosphorylation at these sites has been proposed to disrupt the interaction between H4 and the Na^+, K^+ ATPase cytosolic domain, affecting phospholemman's ability to modulate Na^+, K^+ ATPase function.

To study this, experimenters purified His-tagged Na^+, K^+ ATPase and phospholemman by first solubilizing the membranes with detergent and then using affinity chromatography to isolate the membrane proteins. Activity of the purified pump proteins, both with and without phospholemman, was assayed by measuring ATP consumption under various NaCl concentrations. This assay was repeated using mutants of phospholemman that contain mutations at S68 and S63 to emulate the function of phosphorylated protein (phosphomimetic mutants).

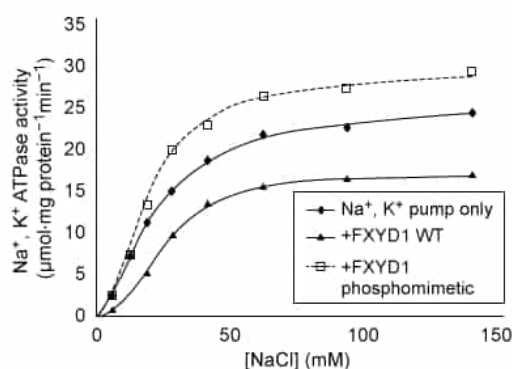


Figure 1. ATPase activity of the Na^+, K^+ pump in detergent micelles was assayed as a function of Na^+ binding. The assay was performed with various concentrations of sodium chloride and a constant concentration of potassium chloride.

What is the main force that stabilizes the structure of H1 of phospholemman?

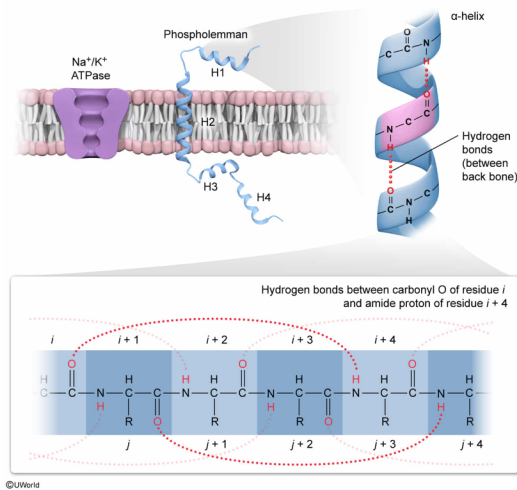
- ☐ A. Hydrophobic interactions that sequester side chains away from water and backbone polar groups away from lipids [26%]
- ☐ B. Covalent bonds between the thiol groups of two residues [4%]
- ☒ C. Hydrogen bonds between amide protons and carbonyl oxygens of non-adjacent amino acid residues [62%]
- ☐ D. Electrostatic interactions between the side chains of distant residues (>10 residues apart) [6%]

Correct

62% answered correctly

Explanation:

Alpha helices are stabilized by hydrogen bonds in the peptide backbone



Secondary structure in proteins is maintained by **hydrogen bonds** between atoms of the **peptide backbone**. **Alpha helices** are one example of protein **secondary structure** that are stabilized by these interactions. Specifically, alpha helix structure is maintained by hydrogen bonding between the **carbonyl oxygen** of a residue at position i (acting as the hydrogen bond acceptor) and the **amide proton** of a residue further down the helix at position $i + 4$ (acting as the hydrogen bond donor). This hydrogen bonding pattern is repeated throughout the structure to form the **right-handed** alpha helix.

The passage states that structure H1 is one of four alpha helices in phospholipid bilayer. As such, this structure is primarily stabilized by hydrogen bonds between amide protons and carbonyl oxygens of nonadjacent residues.

(Choice A) H1 is positioned in the *aqueous* extracellular space and is unlikely to experience strong hydrophobic interactions. This choice would be more descriptive of a transmembrane helix like H2.

(Choice B) Disulfide bonds between two thiol-containing cysteine residues may help to stabilize the **tertiary structure** of proteins, not the secondary structure of an alpha helix.

(Choice D) Interactions between distant side chains is descriptive of the forces that stabilize protein *tertiary* structure, not the secondary structure of an alpha helix.

Educational objective:

Alpha helices are secondary structure elements that appear as right-handed coils of the peptide backbone. Secondary structure is stabilized by hydrogen bonding between peptide backbone amide hydrogens and carbonyl oxygens.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403298

System:
Amino Acids and Proteins

Phospholemman, also known as FXYP1, is a single-pass transmembrane protein that regulates the Na^+, K^+ ATPase pump. Phospholemman consists of four alpha helices: H1 through H4. H1 is located on the extracellular side of the cell. H2 forms the transmembrane region of the protein and interacts intimately with helix TM9 of the Na^+, K^+ ATPase pump. H3 and H4 face the intracellular side of the membrane, where H4 interacts with the cytosolic domain of the Na^+, K^+ ATPase pump through electrostatic interactions. Within H4 exist multiple phosphorylation sites, including S68 and S63, which are phosphorylated by protein kinase A (PKA) and protein kinase C (PKC), respectively. Phosphorylation at these sites has been proposed to disrupt the interaction between H4 and the Na^+, K^+ ATPase cytosolic domain, affecting phospholemman's ability to modulate Na^+, K^+ ATPase function.

To study this, experimenters purified His-tagged Na^+, K^+ ATPase and phospholemman by first solubilizing the membranes with detergent and then using affinity chromatography to isolate the membrane proteins. Activity of the purified pump proteins, both with and without phospholemman, was assayed by measuring ATP consumption under various NaCl concentrations. This assay was repeated using mutants of phospholemman that contain mutations at S68 and S63 to emulate the function of phosphorylated protein (phosphomimetic mutants).

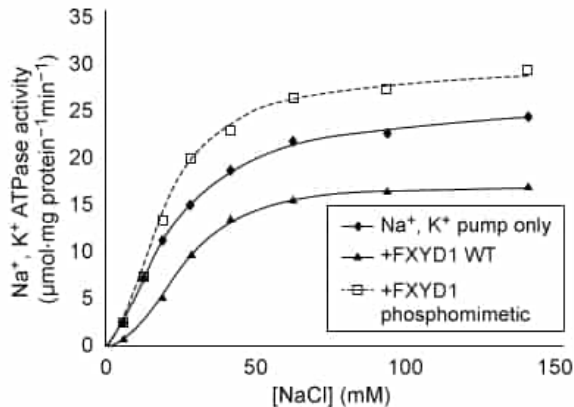


Figure 1. ATPase activity of the Na^+, K^+ pump in detergent micelles was assayed as a function of Na^+ binding. The assay was performed with various concentrations of sodium chloride and a constant concentration of potassium chloride.

Which of the following amino acids is most likely found in H2 of phospholemman?

- ✓ ☒ A. Leucine [79%]
☐ B. Lysine [8%]
☐ C. Glutamine [4%]
☐ D. Serine [7%]

80% answered correctly

Amino acids in transmembrane regions most likely have hydrophobic, nonpolar side chains

Hydrophobic region Hydrophilic region

Nonpolar

$$\begin{array}{c} \text{H} \\ | \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} = \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{CH} \\ / \quad \backslash \\ \text{H}_3\text{C} \quad \text{CH}_3 \end{array}$$

Leucine (L)
Leu

Polar

$$\begin{array}{c} \text{H} \\ | \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} = \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{OH} \end{array}$$

Serine (S)
Ser

Polar

$$\begin{array}{c} \text{H} \\ | \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} = \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{C} = \text{O} \\ | \\ \text{H}_2\text{N} \end{array}$$

Glutamine (Q)
Gln

Charged

$$\begin{array}{c} \text{H} \\ | \\ \text{H}_3\text{N}^+ - \text{C} - \text{C} = \text{O} \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{CH}_2 \\ | \\ \text{+NH}_3 \end{array}$$

Lysine (K)
Lys

©UWorld

The passage states that helix H2 is the transmembrane helix of the **single-pass membrane protein** phospholemman, and therefore H2's side chains are most likely hydrophobic. Of the choices given, three amino acids are polar and would be unstable in the membrane; only leucine, with its $-\text{CH}_2\text{CH}(\text{CH}_3)_2$ side chain, is stable in a hydrophobic environment.

(Choices C and D) Glutamine and serine are polar amino acids with an amide or hydroxyl functional group in their side chains, respectively. Polar side chains are less common in transmembrane helices than nonpolar side chains.

The surface interactions of alpha helices depend on the properties of the amino acid side chains that it comprises. Transmembrane alpha helices in membranes are enriched in amino acids with hydrophobic side chains.

微信Mai ange

Phospholemman, also known as FXYD1, is a single-pass transmembrane protein that regulates the Na^+, K^+ ATPase pump. Phospholemman consists of four alpha helices: H1 through H4. H1 is located on the extracellular side of the cell. H2 forms the transmembrane region of the protein and interacts intimately with helix TM9 of the Na^+, K^+ ATPase pump. H3 and H4 face the intracellular side of the membrane, where H4 interacts with the cytosolic domain of the Na^+, K^+ ATPase pump through electrostatic interactions. Within H4 exist multiple phosphorylation sites, including S68 and S63, which are phosphorylated by protein kinase A (PKA) and protein kinase C (PKC), respectively. Phosphorylation at these sites has been proposed to disrupt the interaction between H4 and the Na^+, K^+ ATPase cytosolic domain, affecting phospholemman's ability to modulate Na^+, K^+ ATPase function.

To study this, experimenters purified His-tagged Na^+, K^+ ATPase and phospholemman by first solubilizing the membranes with detergent and then using affinity chromatography to isolate the membrane proteins. Activity of the purified pump proteins, both with and without phospholemman, was assayed by measuring ATP consumption under various NaCl concentrations. This assay was repeated using mutants of phospholemman that contain mutations at S68 and S63 to emulate the function of phosphorylated protein (phosphomimetic mutants).

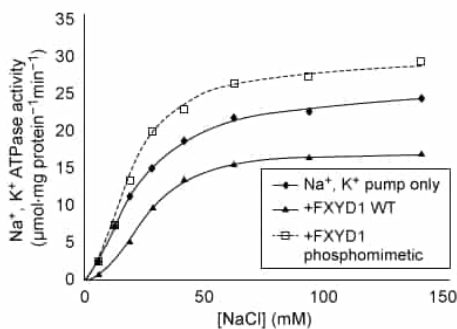


Figure 1. ATPase activity of the Na^+, K^+ pump in detergent micelles was assayed as a function of Na^+ binding. The assay was performed with various concentrations of sodium chloride and a constant concentration of potassium chloride.

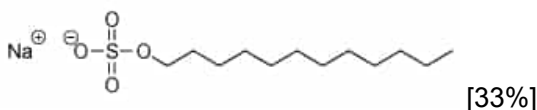
Which compound or mix of compounds was most likely used to elute the His-tagged proteins from the affinity column?

- ☐ A. $\text{Na}^+ \text{Cl}^-$ [21%]
☒ B.



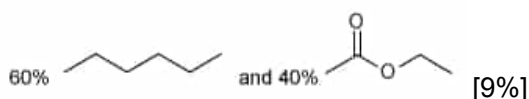
[36%]

- ☐ C.



[33%]

- ☐ D.

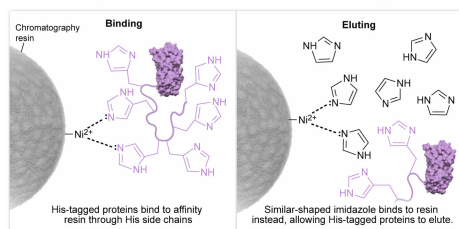


Correct

35% answered correctly

Explanation:

Imidazole competes with histidine side chains to elute His-tagged proteins



Chromatography is a technique used to separate molecules based on their different attractions toward the chromatographic matrix (the stationary phase) and the solvent used to carry the molecules (the mobile phase). **Affinity chromatography** is used commonly in biochemistry and takes advantage of a **specific protein-ligand interaction**, which allows a desired protein to bind specifically and strongly to the column while all other contaminants are washed off. The bound protein can then be gently eluted (removed from the column), often through **competition** by a substance that has been added to the mobile phase.

If a protein does not have a natural ligand that can be easily exploited through chromatography, the protein can instead be genetically modified to possess a **tag** that allows for easy purification. The **His-tag** is a sequence of **histidine** residues that form **coordinate bonds** between their side chains and Ni^{2+} ions that are attached to the chromatography resin. Because the **histidine side chain** is composed of the five-membered aromatic heterocyclic ring **imidazole**, the molecule imidazole by itself can serve to competitively replace histidine interactions with Ni^{2+} . This type of competitive elution allows for the gentle purification of high-purity, still-**functional** protein.

(Choice A) A sodium chloride gradient is used to gently elute proteins in **ion-exchange chromatography**, not in affinity chromatography.

(Choice C) **Sodium dodecyl sulfate (SDS)** is a **denaturant**. While denaturants can disrupt protein structure—and therefore protein-chromatography matrix interactions—they will also render the protein denatured and likely nonfunctional for the future experiments (ie, the activity assays) that the passage states will be done on the purified proteins.

(Choice D) A mixture of hexanes and ethyl acetate is a commonly used mobile phase for normal-phase chromatography of organic molecules. However, these organic solvents will likely denature proteins, rendering them nonfunctional for future experiments.

Educational objective:

Affinity chromatography purifies proteins based on specific protein-ligand interactions. A His-tag can be used for affinity chromatography, allowing proteins to bind to the Ni^{2+} ions through added His residues. The imidazole molecule closely resembles the histidine side chain, disrupting the His- Ni^{2+} interaction and allowing functional protein to elute.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403300
System:
Amino Acids and Proteins

Phospholemman, also known as FXYD1, is a single-pass transmembrane protein that regulates the Na^+, K^+ ATPase pump. Phospholemman consists of four alpha helices: H1 through H4. H1 is located on the extracellular side of the cell. H2 forms the transmembrane region of the protein and interacts intimately with helix TM9 of the Na^+, K^+ ATPase pump. H3 and H4 face the intracellular side of the membrane, where H4 interacts with the cytosolic domain of the Na^+, K^+ ATPase pump through electrostatic interactions. Within H4 exist multiple phosphorylation sites, including S68 and S63, which are phosphorylated by protein kinase A (PKA) and protein kinase C (PKC), respectively. Phosphorylation at these sites has been proposed to disrupt the interaction between H4 and the Na^+, K^+ ATPase cytosolic domain, affecting phospholemman's ability to modulate Na^+, K^+ ATPase function.

To study this, experimenters purified His-tagged Na^+, K^+ ATPase and phospholemman by first solubilizing the membranes with detergent and then using affinity chromatography to isolate the membrane proteins. Activity of the purified pump proteins, both with and without phospholemman, was assayed by measuring ATP consumption under various NaCl concentrations. This assay was repeated using mutants of phospholemman that contain mutations at S68 and S63 to emulate the function of phosphorylated protein (phosphomimetic mutants).

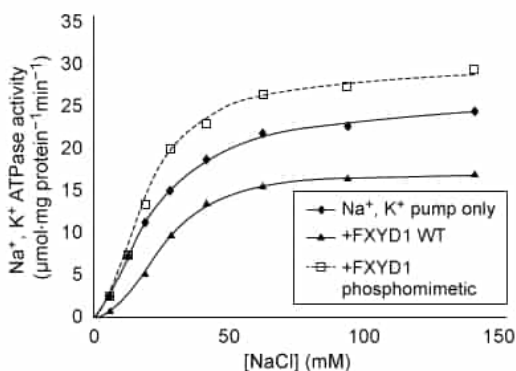


Figure 1. ATPase activity of the Na^+, K^+ pump in detergent micelles was assayed as a function of Na^+ binding. The assay was performed with various concentrations of sodium chloride and a constant concentration of potassium chloride.

Based on the passage, how do treatment with PKA and PKC affect the isoelectric point of phospholemman?

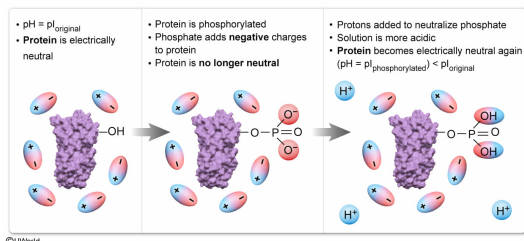
- ☐ A. PKA decreases the isoelectric point, whereas PKC increases the isoelectric point. [1%]
- ☐ B. PKA increases the isoelectric point, whereas PKC decreases the isoelectric point. [3%]
- ☒ C. Both PKA and PKC decrease the isoelectric point. [63%]
- ☐ D. Both PKA and PKC increase the isoelectric point. [30%]

Correct

64% answered correctly

Explanation:

Phosphorylation decreases the isoelectric point (pI) of proteins



Proteins consist of chains of amino acids with varied **side chains**. Acidic and basic side chains are ionizable and can either gain or lose protons in solution, thus imparting an ionic charge to itself and to the whole protein. The overall net ionic charge of the protein depends on the balance of acidic and basic side chains, as well as on the pH of the solution the protein is in. The **isoelectric point (pI)** of a protein is the pH at which the protein is electrically neutral, with a net charge of ± 0 .

Phosphorylation, either by PKA or by PKC, will add a phosphate group that can be deprotonated to have **negative charges**. When the pH equals the pI of the unphosphorylated protein, the unphosphorylated form is electrically neutral. However, at the same pH, the phosphorylated protein will now be slightly negative. To neutralize this overall negative charge, positive charges must be added by adding protons to the solution, some of which bind to the protein. This acidifies the solution, lowering the pH needed to achieve a neutral protein, and thus lowering the pI. Note that it is not necessarily the phosphate group itself that must be protonated, but *any* functional group on the protein that can accept a proton to neutralize the *overall* negative charge.

(Choices A and B) Though activated by different pathways, PKA and PKC are both kinases that will add phosphate groups to phospholemman. Regardless of enzyme, the added phosphates are both negatively charged and will decrease the pI.

(Choice D) Phosphorylation *decreases* the pI, rather than increasing it.

Educational objective:

Phosphorylation is a post-translational modification of proteins characterized by the addition of negatively charged phosphate groups to the protein. When assessing a protein's isoelectric point (pI), these negative charges must be neutralized by the addition of acidic protons, thus lowering the pI relative to unphosphorylated protein.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403301

System:
Amino Acids and Proteins

Phospholemman, also known as FXYD1, is a single-pass transmembrane protein that regulates the Na^+, K^+ ATPase pump. Phospholemman consists of four alpha helices: H1 through H4. H1 is located on the extracellular side of the cell. H2 forms the transmembrane region of the protein and interacts intimately with helix TM9 of the Na^+, K^+ ATPase pump. H3 and H4 face the intracellular side of the membrane, where H4 interacts with the cytosolic domain of the Na^+, K^+ ATPase pump through electrostatic interactions. Within H4 exist multiple phosphorylation sites, including S68 and S63, which are phosphorylated by protein kinase A (PKA) and protein kinase C (PKC), respectively. Phosphorylation at these sites has been proposed to disrupt the interaction between H4 and the Na^+, K^+ ATPase cytosolic domain, affecting phospholemman's ability to modulate Na^+, K^+ ATPase function.

To study this, experimenters purified His-tagged Na^+, K^+ ATPase and phospholemman by first solubilizing the membranes with detergent and then using affinity chromatography to isolate the membrane proteins. Activity of the purified pump proteins, both with and without phospholemman, was assayed by measuring ATP consumption under various NaCl concentrations. This assay was repeated using mutants of phospholemman that contain mutations at S68 and S63 to emulate the function of phosphorylated protein (phosphomimetic mutants).

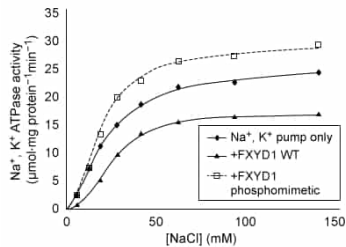


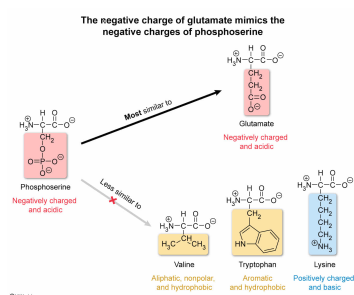
Figure 1. ATPase activity of the Na^+, K^+ pump in detergent micelles was assayed as a function of Na^+ binding. The assay was performed with various concentrations of sodium chloride and a constant concentration of potassium chloride.

Which of the following amino acids was most likely used to substitute for S68 and S63 when designing the phosphomimetic mutants?

- ☐ A. Val [11%]
- ☐ B. Lys [12%]
- ☐ C. Trp [11%]
- ☒ D. Glu [63%]

Correct
64% answered correctly

Explanation:



Amino acids, the building blocks of proteins, differ from each other through the chemical structure and properties of their **side chains**. In evolutionary biology as well as in laboratory biochemistry, mutations in the identity of an amino acid can lead to changes in function of the protein resulting from the change in properties of the amino acid side chain. If the properties before and after the mutation are similar, then a mutant protein can arise with similar (though not necessarily identical) properties to the wild type. This is known as a **conservative mutation**.

The passage states that the researchers wanted to create phosphomimetic mutants whose **function was similar to** phosphorylated phospholemman. Thus, the most appropriate choice is to replace serine with another amino acid with similar properties to phosphorylated serine (ie, phosphoserine). With the attached phosphate group, phosphoserine becomes a **negatively charged, acidic amino acid**. Of the choices given, only glutamate (Glu) has the properties of being a **negatively charged, acidic amino acid**. This conservation of charge allows glutamate to disrupt the electrostatic interaction between H4 and Na^+, K^+ ATPase in a manner similar to that of phosphoserine in wild-type, phosphorylated phospholemman.

(Choice A) Valine is an aliphatic, nonpolar, and hydrophobic amino acid, unlike the negatively charged phosphoserine.

(Choice B) Lysine is a *positively* charged, basic amino acid. With the opposite charge of phosphoserine, lysine is more likely to *enhance* the attraction between H4 and Na^+, K^+ ATPase rather than disrupt it.

(Choice C) Tryptophan is a large, bulky, aromatic amino acid, unlike the negatively charged phosphoserine.

Educational objective:

Amino acids with similar side chain properties can often substitute for each other while leaving protein function intact. Because glutamate and phosphoserine both have negatively charged, acidic residues, glutamate is often a viable choice to effectively mimic protein phosphorylation.

Time spent:0 sec**Subject:**
Biochemistry**QID:**403302**System:**
Amino Acids and Proteins

Phospholemman, also known as FXYD1, is a single-pass transmembrane protein that regulates the Na^+, K^+ ATPase pump. Phospholemman consists of four alpha helices: H1 through H4. H1 is located on the extracellular side of the cell. H2 forms the transmembrane region of the protein and interacts intimately with helix TM9 of the Na^+, K^+ ATPase pump. H3 and H4 face the intracellular side of the membrane, where H4 interacts with the cytosolic domain of the Na^+, K^+ ATPase pump through electrostatic interactions. Within H4 exist multiple phosphorylation sites, including S68 and S63, which are phosphorylated by protein kinase A (PKA) and protein kinase C (PKC), respectively. Phosphorylation at these sites has been proposed to disrupt the interaction between H4 and the Na^+, K^+ ATPase cytosolic domain, affecting phospholemman's ability to modulate Na^+, K^+ ATPase function.

To study this, experimenters purified His-tagged Na^+, K^+ ATPase and phospholemman by first solubilizing the membranes with detergent and then using affinity chromatography to isolate the membrane proteins. Activity of the purified pump proteins, both with and without phospholemman, was assayed by measuring ATP consumption under various NaCl concentrations. This assay was repeated using mutants of phospholemman that contain mutations at S68 and S63 to emulate the function of phosphorylated protein (phosphomimetic mutants).

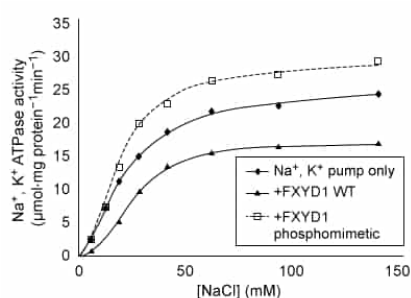


Figure 1. ATPase activity of the Na^+, K^+ pump in detergent micelles was assayed as a function of Na^+ binding. The assay was performed with various concentrations of sodium chloride and a constant concentration of potassium chloride.

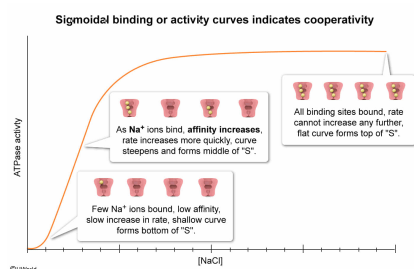
Which statement is best supported based on the passage and the data in figure 1?

- ☐ A. Na^+, K^+ ATPase activity is independent of potassium ion concentration. [9%]
- ☒ B. The first, second, and third sodium ions bind to the Na^+, K^+ ATPase with different affinities. [28%]
- ☐ C. Phosphorylation causes complete dissociation of Na^+, K^+ ATPase from phospholemman. [21%]
- ☐ D. Formation of an electrochemical gradient across the membrane slows Na^+, K^+ ATPase over the course of the experiment. [40%]

Correct

29% answered correctly

Explanation:



In proteins, **cooperativity** occurs if the binding of one ligand influences the binding of subsequent ligands. When binding at one site makes binding of subsequent ligands *easier*, the interaction is an example of **positive cooperativity**. Positive cooperativity can be recognized by an S-shaped **sigmoidal curve** on a plot of activity versus ligand concentration. The curve starts shallow, indicating low-affinity binding at low ligand concentrations. The curve steepens as **ligand-induced conformational changes** cause **high-affinity binding** for subsequent ligands. The curve flattens again as proteins become saturated with ligand, forming the top of the sigmoidal curve.

The Na^+, K^+ ATPase, also known as the **sodium-potassium pump**, is a membrane protein that hydrolyzes one molecule of ATP to transport three sodium ions to the extracellular space and bring two potassium ions into the cytosol. Because these

three actions are linked, the rate of ATP hydrolysis depends upon the protein's binding of sodium ions and therefore on the sodium ion concentration. Further, because Na^+, K^+ ATPase binds and transports *three* sodium ions, this opens the possibility of cooperativity. Figure 1 shows that Na^+, K^+ ATPase displays sigmoidal activity curves, confirming positive cooperativity as each successive sodium ion binds with increasing affinity.

(Choice A) In the experiments shown, potassium ion concentration was held constant as a standardized variable. Because no data were shown with varying potassium ion concentrations, no conclusions can be drawn regarding the relationship between activity and K^+ concentration.

(Choice C) If phospholemman and Na^+, K^+ -ATPase were completely dissociated upon phosphorylation, Na^+, K^+ -ATPase activity would be expected to return to control levels (Na^+, K^+ -ATPase alone). Instead, phosphorylation induces higher Na^+, K^+ -ATPase activity, indicating that phosphorylated phospholemman acts as a positive modulator.

(Choice D) The experiments were performed in protein-detergent micelles without an intact membrane. Without the membrane to separate compartments, the Na^+, K^+ -ATPase can continue hydrolyzing ATP but cannot generate an electrochemical gradient across a membrane.

Educational objective:

In proteins, cooperativity is characterized by the ligand affinity changing as more ligands bind. Positive cooperativity, which occurs when subsequent ligands bind with stronger affinity than the first, can be identified by a sigmoidal curve on a plot of activity versus ligand concentration.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403303
System:
Amino Acids and Proteins